MLP Hidden States Change Progressively Across Depth Instead of Becoming Unrelated

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Abstract

Representation similarity methods are widely used to analyze deep networks, but controlled studies on plain multilayer perceptrons (MLPs) remain less common than analogous work on convolutional or attention-based models. We ask a simple question: are hidden states in different MLP layers effectively unrelated, or do they preserve enough shared structure that layer-wise comparison is meaningful? To answer this, we train matched four-hidden-layer ReLU MLPs on MNIST, FASHION-MNIST, and CIFAR-10 using three random seeds and compare hidden activations with linear CKA, SVCCA, PWCCA, and sample-wise cosine similarity. We complement these geometric comparisons with linear probes on every hidden layer.

Our results show a clear within-model locality pattern. Adjacent layers are more similar than non-adjacent layers in all 12 dataset-metric comparisons, with 10 reaching $p < 0.05$. Cross-seed comparisons are weaker: same-depth layers exceed off-depth layers in 8 of 12 cases, and only 5 are statistically significant. Linear probes show that later layers usually improve decodable task information, from 0.8767 to 0.9108 on MNIST and from 0.8308 to 0.8450 on FASHION-MNIST, while CIFAR-10 peaks at layer 3 with 0.4292.

These findings argue against the view that MLP hidden spaces are completely rewritten at each layer. Instead, plain feedforward MLPs appear to transform representations progressively, although the strength of this conclusion depends on which similarity metric is used.

1 Introduction

Hidden-state similarity is not a side question. Interpretability, pruning, model surgery, and transfer arguments all become easier to justify if intermediate layers refine a shared representation instead of rebuilding a new feature basis from scratch at every step. Our main question is therefore direct: when a plain multilayer perceptron processes an input through depth, do its hidden states become totally different, or do they remain comparable in measurable ways?

This question matters because many practical analyses assume some form of continuity across depth. If adjacent layers preserve substantial shared structure, then layer-wise comparisons, lightweight editing, and partial model reuse have a stronger empirical foundation. If they do not, then similarity-based claims about representational progression become much harder to defend.

Existing tools for representation comparison are mature. SVCCA studies compare dominant subspaces across layers and checkpoints Raghu et al. [2017]. Linear CKA was proposed in part because correspondence tests exposed failure modes for CCA-style methods in high-dimensional settings Kornblith et al. [2019]. Later work broadened the picture by separating geometric similarity from functional compatibility Csiszárik et al. [2021], questioning how reliable CKA is under distributional shifts and feature manipulations Davari et al. [2023], and clarifying what common similarity measures imply about linear decoders Harvey et al. [2024]. However, the recent empirical empha-

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sis has been stronger for convolutional networks and transformers than for controlled plain-MLP studies.

We focus on that gap. We train the same four-hidden-layer ReLU MLP on three standard image benchmarks of increasing difficulty, compare hidden activations with four complementary metrics, and add linear probes as a task-information sanity check. This design lets us separate three questions: whether nearby layers are geometrically closer than distant ones, whether the same layer role recurs across random seeds, and whether deeper layers improve linearly decodable information even when they remain similar to earlier layers.

What do we find? The answer is consistent within models and mixed across seeds. Adjacent layers are more similar than non-adjacent layers in all 12 dataset-metric comparisons, with 10 significant at $p < 0.05$. Same-depth cross-seed similarity beats off-depth similarity in only 8 of 12 cases. Linear probes improve with depth on MNIST and FASHION-MNIST, and peak at layer 3 on CIFAR-10. Taken together, these results support a progressive-transformation view of MLP depth rather than a complete rewriting view.

Our contributions are:

- We conduct a controlled multi-metric benchmark of hidden-state similarity in plain feed-forward MLPs on MNIST, FASHION-MNIST, and CIFAR-10.
- We compare within-model layer locality and cross-seed same-depth recurrence, which separates representational continuity from seed-specific alignment.
- We complement geometric similarity scores with linear probes, showing that later layers can improve decodable task information while remaining substantially similar to earlier ones.
- We provide a concrete empirical answer: MLP hidden states are not totally different across depth, but the strength of that claim depends on the metric and is strongest within a single trained model.

The rest of the paper is organized as follows. Section 2 situates the study in prior work on representation similarity. Section 3 describes the training setup, similarity measures, and statistical comparisons. Section 4 reports the main empirical results, and Section 5 discusses interpretation, limitations, and broader implications.

2 Related Work

Subspace-based representation comparison. Early work on neural representation similarity focused on subspace alignment. SVCCA removes low-variance directions and then applies canonical correlation analysis to compare hidden representations across layers, checkpoints, and models Raghu et al. [2017]. PWCCA extends this line by reweighting canonical directions according to their contribution to the representation, which can make the score better reflect dominant activation structure. These methods are natural starting points for our setting because they compare whole hidden spaces rather than individual neurons. Our study uses both SVCCA and PWCCA, but evaluates them in a simple MLP regime where architecture and training are tightly controlled.

Kernel-style similarity and metric reliability. Kornblith et al. [2019] argued that linear CKA is often more reliable than CCA-style measures for high-dimensional correspondence tests, making it a strong default baseline for layer-by-layer analysis. At the same time, later work showed that no single geometric metric is sufficient. Davari et al. [2023] demonstrated that CKA can be sensitive to outliers and certain feature transformations even when downstream usefulness is preserved. The broader survey of Klabunde et al. [2023] makes the same point at a higher level: different similarity scores encode different invariances and answer different scientific questions. We follow this guidance directly by reporting four metrics rather than relying on one preferred score.

Functional compatibility and decodable information. Geometric similarity is not the same as functional interchangeability. Model stitching work showed that representations can be functionally compatible even when their geometric similarity is modest Csiszárik et al. [2021]. More recent theoretical work linked common similarity measures to alignment between optimal linear readouts, clarifying when geometric agreement should imply agreement in decodable information Harvey

Table 1: Test accuracy across three random seeds. Higher is better. Best mean accuracy is in **bold**.

DATASET	Mean	Std	Min	Max
CIFAR-10	0.3937	0.0033	0.3900	0.3965
FASHION-MNIST	0.8228	0.0098	0.8120	0.8310
MNIST	**0.9270**	0.0039	0.9245	0.9315

et al. [2024]. Recent layer-wise analyses also report that nearby layers tend to be more similar than distant ones, especially in larger modern architectures Jiang et al. [2024]. Our paper positions itself as a controlled benchmark for plain MLPs that brings these threads together: we measure geometry with multiple metrics, test layer recurrence across seeds, and add linear probes to track task information by depth.

3 Methodology

Research question and hypotheses. We ask whether hidden-state spaces in different layers of a trained MLP are largely shared rather than totally different, and if they are shared, what changes measurably across depth. We test four hypotheses: adjacent layers should be more similar than distant layers; this distance effect should appear across multiple metrics; the same hidden depth should recur more strongly across seeds than mismatched depths; and later layers should improve linear decodability without collapsing into unrelated spaces.

Datasets and preprocessing. We use MNIST, FASHION-MNIST, and CIFAR-10, loaded from local Hugging Face `save_to_disk` artifacts. Because the installed PyTorch build could not use the available GPUs, we ran CPU-feasible stratified subsets: 12,000 original training examples and 2,000 test examples for MNIST and FASHION-MNIST, and 15,000 training plus 2,000 test examples for CIFAR-10. We flattened images into vectors, scaled pixel values to $[0, 1]$, and split the training subset into stratified 85/15 train/validation partitions. The resulting train/validation/test shapes are $10{,}200/1{,}800/2{,}000$ for MNIST and FASHION-MNIST, and $12{,}750/2{,}250/2{,}000$ for CIFAR-10. All datasets had zero missing values.

Model and optimization. For every dataset we trained the same four-hidden-layer ReLU MLP with hidden width 128, dropout, AdamW optimization, batch size 512, early stopping on validation accuracy, and deterministic seeding. We used three random seeds: 42, 123, and 456. The software environment was Python $3.12.8$ with PyTorch $2.11.0+$`cu130`. Four RTX A6000 GPUs were visible to the host, but `torch.cuda.is_available()` was false, so all training and evaluation ran on CPU.

Representation similarity metrics. We compare hidden activations with four metrics chosen to cover different invariances. Linear CKA measures similarity between centered feature kernels Kornblith et al. [2019]. SVCCA and PWCCA compare dominant subspaces and canonical directions Raghu et al. [2017]. Sample-wise cosine similarity provides a simpler activation-level baseline. Using this set reduces the chance that a single metric artifact drives the main conclusion.

Comparison design. We evaluate two kinds of alignment. First, within each trained model we compare all hidden-layer pairs and aggregate them into adjacent pairs $(1, 2)$, $(2, 3)$, $(3, 4)$ and non-adjacent pairs $(1, 3)$, $(1, 4)$, $(2, 4)$. This tests whether similarity declines with layer distance. Second, across seeds we compare same-depth pairs against off-depth pairs to test whether layer roles recur across independent initializations. We report mean differences, permutation-test p-values, and Cohen's d from the result summary.

Linear probes. To track task information by depth, we train a linear classifier on each hidden layer and measure held-out accuracy. These probes do not claim full functional equivalence, but they provide a direct sanity check on whether deeper layers preserve or improve decodable label information.

4 Results

Base task performance. The shared architecture reaches mean test accuracy 0.9270 on MNIST, 0.8228 on FASHION-MNIST, and 0.3937 on CIFAR-10, which establishes a simple but non-trivial

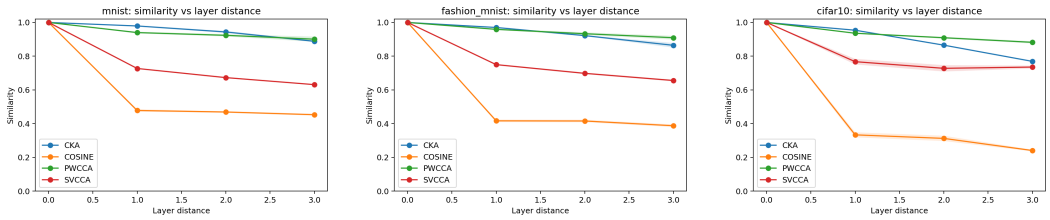

Figure 1: Within-model similarity declines with layer distance across all three datasets. The same locality pattern appears with different absolute scales, with the cleanest separation under CKA, SVCCA, and PWCCA.

Table 2: Adjacent-layer versus non-adjacent-layer similarity within the same trained model. Positive differences mean nearby layers are more similar. The largest positive difference for each dataset is in **bold**.

DATASET	METRIC	Adjacent Mean	Non-Adjacent Mean	Difference	p-value	Cohen's d
CIFAR-10	CKA	**0.9540**	0.8335	**0.1206**	0.0002	2.43
CIFAR-10	Cosine	0.3334	0.2885	0.0449	0.0048	0.75
CIFAR-10	PWCCA	0.9361	0.9004	0.0357	0.0002	1.99
CIFAR-10	SVCCA	0.7674	0.7305	0.0369	0.0322	0.56
FASHION-MNIST	CKA	**0.9703**	0.9028	**0.0674**	0.0002	2.09
FASHION-MNIST	Cosine	0.4166	0.4063	0.0103	0.2280	0.30
FASHION-MNIST	PWCCA	0.9591	0.9254	0.0338	0.0002	1.23
FASHION-MNIST	SVCCA	0.7499	0.6841	0.0658	0.0002	2.37
MNIST	CKA	0.9793	0.9253	0.0540	0.0002	1.85
MNIST	Cosine	0.4780	0.4636	0.0144	0.0634	0.46
MNIST	PWCCA	0.9400	0.9161	0.0239	0.0002	1.22
MNIST	SVCCA	**0.7271**	0.6591	**0.0680**	0.0002	3.18

benchmark across task difficulty. Best-seed performance follows the same ranking, with MNIST easiest and CIFAR-10 hardest.

Within-model locality is the dominant result. Table 2 shows that adjacent layers are more similar than non-adjacent layers in all 12 dataset-metric comparisons. The effect is strongest and most stable for CKA, SVCCA, and PWCCA. For example, on CIFAR-10, adjacent-layer CKA exceeds non-adjacent CKA by 0.1206 with $p = 0.0002$ and Cohen's $d = 2.43$. The same qualitative pattern appears on MNIST and FASHION-MNIST. Cosine similarity is directionally consistent everywhere, but its weaker margins on MNIST and FASHION-MNIST do not reach $p < 0.05$.

Cross-seed layer recurrence is weaker. Table 3 reports same-depth versus off-depth similarity across random seeds. Here the story is mixed: 8 of 12 comparisons favor same-depth alignment, but only 5 are statistically significant. CKA remains the most consistent metric, showing positive and significant same-depth gaps on all three datasets. By contrast, cosine is inconsistent, and both PWCCA and SVCCA become weak or negative on CIFAR-10. This means that the claim "layers are comparable" is empirically stronger within a single model than across independent initializations.

Later layers improve linear decodability. Figure 2 complements the geometric results with a task-information view. On MNIST, mean probe accuracy rises from 0.8767 at layer 1 to 0.9108 at layer 4. On FASHION-MNIST, the best mean accuracy is 0.8450 at layers 3 and 4, up from 0.8308 at layer 1. On CIFAR-10, the best layer is layer 3 with 0.4292, compared with 0.4100 at layer 1. These gains are modest compared with the overall class difficulty, but they are consistent with a refinement view of depth: later layers improve linearly decodable information while still remaining similar to earlier layers.

Ablation by metric. The comparison across metrics is itself an informative ablation. CKA, SVCCA, and PWCCA all support the progressive-transformation picture within models, while cosine provides the same direction with less stability. Across seeds, only CKA remains consis-

Table 3: Same-depth versus off-depth similarity across random seeds. Positive differences mean the same hidden depth recurs across seeds more strongly than mismatched depths. The strongest positive difference for each dataset is in **bold**.

DATASET	METRIC	Same-Depth Mean	Off-Depth Mean	Difference	p-value	Cohen's d
CIFAR-10	CKA	0.9359	0.8576	**0.0783**	0.0008	1.41
CIFAR-10	Cosine	0.3211	0.3020	0.0191	0.4993	0.19
CIFAR-10	PWCCA	0.8878	0.8887	-0.0010	0.6237	-0.14
CIFAR-10	SVCCA	0.6715	0.7018	-0.0303	0.1194	-0.48
FASHION-MNIST	CKA	0.9160	0.8777	**0.0383**	0.0378	0.73
FASHION-MNIST	Cosine	0.4051	0.4137	-0.0086	0.4265	-0.24
FASHION-MNIST	PWCCA	0.9411	0.9254	0.0158	0.1400	0.47
FASHION-MNIST	SVCCA	0.6954	0.6761	0.0193	0.0018	0.87
MNIST	CKA	0.9560	0.9234	**0.0326**	0.0014	0.84
MNIST	Cosine	0.4562	0.4696	-0.0134	0.4579	-0.20
MNIST	PWCCA	0.9166	0.9078	0.0089	0.3583	0.26
MNIST	SVCCA	0.6818	0.6568	0.0250	0.0014	0.98

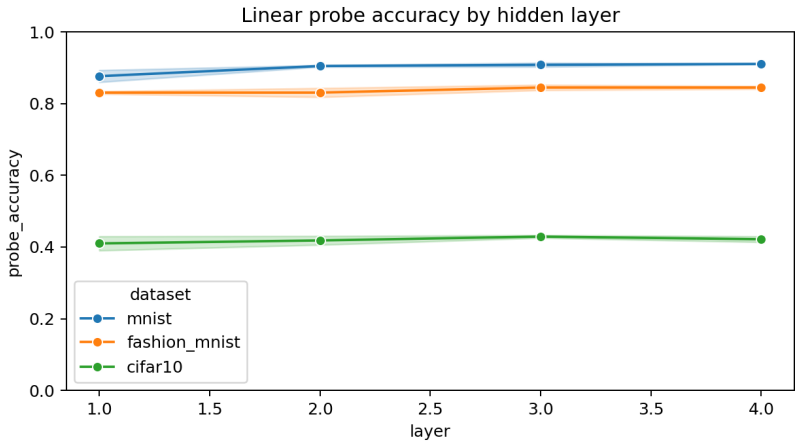

Figure 2: Linear probe accuracy by hidden layer. Later layers usually improve decodable task information, although the strongest gain on CIFAR-10 appears at layer 3 rather than layer 4.

tently positive across datasets. This pattern suggests that metric choice is not a cosmetic decision; it changes how strongly one should believe claims about recurring layer roles across training runs.

5 Discussion

What do the results imply? The central empirical answer is negative to the strongest form of the null hypothesis: hidden states in different MLP layers are not totally different. The clearest evidence is local. Adjacent layers beat non-adjacent layers in every within-model comparison, often by large margins under CKA, SVCCA, and PWCCA. At the same time, deeper layers improve probe accuracy, which means similarity is not merely redundancy. The more plausible interpretation is that depth progressively refines a shared representational space.

Why are cross-seed results weaker? Cross-seed alignment asks more than continuity. It asks whether independently trained models reuse the same depth-specific organization. The mixed results suggest that a layer can play a similar functional role without preserving one stable geometric template across seeds. This is especially clear on CIFAR-10, where PWCCA and SVCCA become flat or negative despite positive within-model locality. For practical interpretability, this means layer-to-layer comparison is safer inside one model than across separately trained replicas.

Limitations. This is a controlled study of plain feedforward MLPs on flattened image inputs. The findings should not be transferred directly to architectures with convolution, residual pathways, or attention. The runtime environment also forced CPU execution and moderate dataset subsampling despite four visible GPUs, which limits scale and may suppress stronger convergence on CIFAR-10. Finally, every similarity score compresses rich geometry into a scalar summary, and SVCCA/PWCCA can be numerically sensitive.

Broader implications. Even with those limitations, the study provides a useful empirical baseline. It supports the idea that layer-wise analysis in simple feedforward networks is meaningful, and it shows why multi-metric reporting is necessary. The most direct next extension is functional: layer stitching could test whether the observed geometric continuity translates into interchangeable computation. More broadly, the same benchmark design can be used to study when progressive similarity breaks down as width, depth, or architecture changes.

6 Conclusion

We presented a controlled benchmark of hidden-state similarity in plain four-layer MLPs trained on MNIST, FASHION-MNIST, and CIFAR-10. By combining linear CKA, SVCCA, PWCCA, cosine similarity, and per-layer linear probes, we separated within-model layer locality from cross-seed recurrence and from task-information growth with depth.

Our main finding is simple: MLP hidden spaces across depth are not totally different. Adjacent layers are consistently more similar than distant ones, while later layers usually improve decodable label information. Cross-seed comparisons are weaker, which shows that representational continuity within a model is easier to establish than stable layer identity across random initializations.

The next steps are straightforward. Functional stitching can test whether these similarities support interchangeable computation. Additional sweeps over width, depth, and pre- versus post-activation representations can identify when progressive similarity stops holding. Extending the same protocol to residual networks and transformers would clarify how architecture changes the geometry of layer-wise progression.

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